

1)

i) (about) 3.5 billion/ (about) 3.5×10^9 [1]

ii)

a) Sum of all chemical activities / catabolic and anabolic reactions / taking place in an organism. [1]

b) Irreversible increase in dry mass / weight (of an organism) [1]

c) Irreversible changes that occur during the life span (of an organism) [1]

iii)

a) Production of high yielding varieties (of plants and animals) . Production of disease resistant varieties (of plants and animals). Improving the postharvest technologies/ methods. [3]

b) Increase of (growth rate of) human population [1]

iv) Archaeon [1]

v)

a) Paleozoic [1]

b) Mesozoic [1]

c) Paleozoic [1]